

PRAKHAR SHUKLA

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PROFESSIONAL SUMMARY

Solutions-oriented Software Engineer and Cloud Specialist with expertise in designing scalable backend systems, architecting highly available AWS infrastructure, and implementing modern DevOps CI/CD pipelines. Proficient in integrating advanced AI workflows and Large Language Models (LLMs) into production-ready applications. Passionate about driving technical excellence, optimizing system performance, and delivering robust, user-centric software solutions.

Portfolio

TECHNICAL SKILLS

AWS & Cloud: Lambda, EC2, ECS, EKS, S3, IAM, RDS, VPC, CloudFormation, GCP

Programming: Java, Python, JavaScript, TypeScript, C++

DevOps & CI/CD: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

AI & ML Integration: LangChain, OpenAI APIs, Vector Databases, Prompt Engineering

Architecture & Design: Microservices, REST APIs, Redis, Nginx, Kafka, Load Balancing

Web Frameworks: Node.js, Express.js, React.js, FastAPI

Databases: MySQL, PostgreSQL, MongoDB, DynamoDB

Tools & OS: Git, AWS Console, Postman, Linux, Ubuntu, Windows Server

PROFESSIONAL EXPERIENCE

Ethara AI | LLM Post Training Intern

Mar 2026 – May 2026

- Engineered LLM post-training evaluation pipelines, optimizing prompt structures to drastically improve user interaction and instruction adherence.
- Conducted rigorous data analysis on AI-generated outputs, identifying edge cases and contributing to a 15% measurable increase in overall model performance.
- Spearheaded the integration of systematic feedback loops and advanced prompt engineering techniques to ensure highly factual and context-aware AI responses.

ENGINEERING PROJECTS

MULTI-SERVER WEB HOSTING INFRASTRUCTURE

- Architected a highly available, multi-tier cloud infrastructure on AWS leveraging EC2, Auto Scaling Groups, and Linux servers to efficiently manage fluctuating web traffic.
- Secured network architecture by implementing precise VPC subnets, Security Groups, and IAM roles, ensuring strict access control across all resources.
- Integrated Amazon RDS for relational data persistence and S3 combined with CloudFront CDN to accelerate global content delivery by caching static assets.

AI AGENT WORKFLOW AUTOMATION PLATFORM | [GitHub](#)

- Designed and developed a scalable workflow builder empowering users to visually map and connect multi-agent AI execution pipelines.
- Integrated LangChain and advanced LLM reasoning capabilities to automate complex, multi-step tasks natively within the platform.
- Engineered robust state management and real-time execution tracking architectures using Redis and MongoDB to maintain data integrity across asynchronous agent operations.

FULL-STACK TASK MANAGEMENT SYSTEM | [GitHub](#)

- Engineered a scalable RESTful API backend using Node.js and Express, facilitating comprehensive task operations and user management.
- Implemented highly secure, role-based access control (RBAC) and JWT authentication mechanisms to isolate administrative and user environments.
- Deployed the full-stack containerized application to AWS, establishing automated CI/CD pipelines and rigorous security configurations.

POSCO E-COMMERCE FOOD DELIVERY PLATFORM | [GitHub](#)

- Built a responsive, dynamic front-end interface optimized for seamless menu navigation, cart state management, and mobile accessibility.
- Architected a NoSQL database schema using MongoDB to efficiently handle dynamic restaurant data, complex orders, and user transactional history.
- Engineered rapid REST API endpoints for secure user authentication, order processing, and streamlined checkout flows.

CERTIFICATIONS

• AWS Cloud Practitioner Essentials	[View Certificate]
• AWS Foundations of Prompt Engineering	[View Certificate]
• AWS Getting Started With IoT	[View Certificate]
• Cisco – Switching, Routing and Wireless Essentials	[View Certificate]
• Cisco - Introduction to Networks	[View Certificate]

EDUCATION

GLA University Mathura <i>B.Tech in Computer Science & Engineering</i> <i>CGPA: 7.65</i>	2022 – 2026
Lucknow Public School <i>Class XII (Intermediate)</i> <i>Madhoganj, Hardoi</i>	2022
Lucknow Public School <i>Class X (High School)</i> <i>Madhoganj, Hardoi</i>	2020

LEADERSHIP & EXTRACURRICULARS

- General Secretary, Computer Science & Engineering Club (2023–2025) – Directed technical events, workshops, and peer mentoring sessions, fostering a collaborative engineering culture.

Relevant Coursework (2022–2026):

- Data Science and Analytics
- IIoT communication and standard interface
- Introduction to Internet of Things
- Machine learning for IoT
- Smart Industrial Connectivity
- Artificial Intelligence and Machine Learning